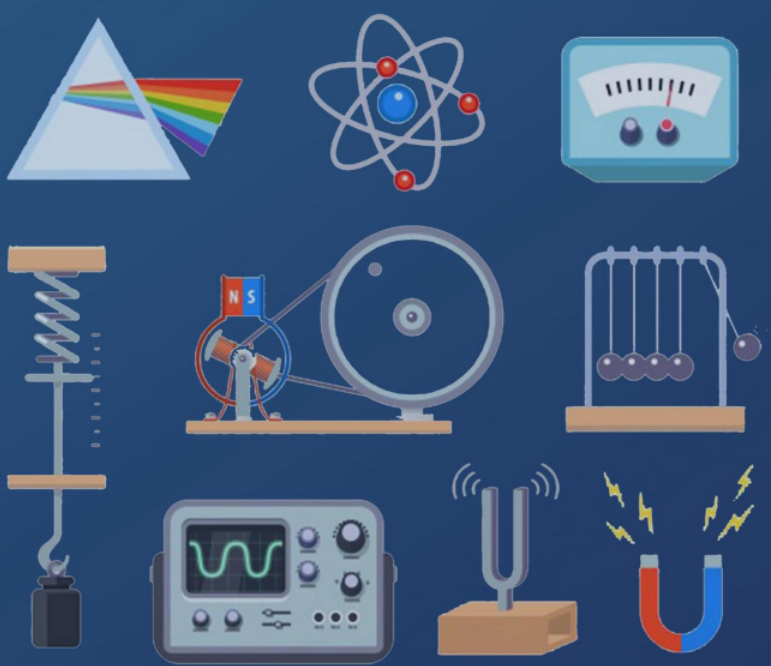
PHYSICS

Vector



SHEET

eSaral.com

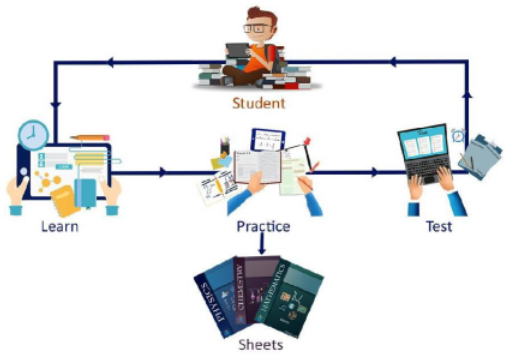
Contents

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Introduction of Vectors	1, 2	1		
2	Addition of Vectors	3, 4	2, 9		
3	Parallelogram Rule of Vector Addition	5 to 8	3 to 6	1, 2	1, 3, 4, 2
4	Unit Vector	9 to 11			
5	Resolution of Vectors	12, 13	10	3	
6	Expressing Vectors in Unit Vector Notation	14 to 16	7, 11		
7	Position and Displacement Vector	17, 18	12	4 to 6	5 to 7
8	Scalar Product of Two Vectors	19 to 21	8	7	8

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction of Vectors

Q1. Identify the scalar quantity :

(A) Force (B) Acceleration

(C) Displacement (D) Work

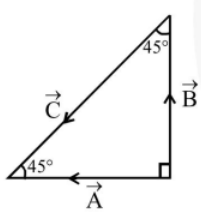
Q2. I started walking down a road in morning facing the sun. After walking for some–time, I turned to my left, then I turned to the right once again. In which direction was I going then :

(A) East (B) North–west

(C) North–east (D) South

Addition and Subtraction of Vectors

Q3. The correct relation for the triangle shown in figure :-



(A) $\vec{A}+\vec{B}=\vec{C}$ (B) $\vec{A}+\vec{B}+\vec{C}=0$

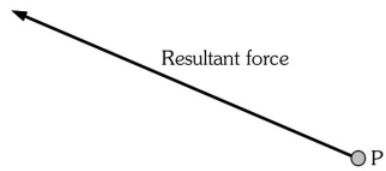
(C) $\vec{A}+\vec{C}=\vec{B}$ (D) $\vec{B}+\vec{C}-\vec{A}=0$

Q4. A man moves in an open field such that after moving 10 m in a straight line, he makes a sharp turn of 60° to his left. The total displacement of the man just after 7 such turns is -

(A) 10 m (B) 20 m

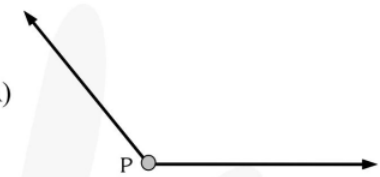
(C) 70 m (D) 30 m

Q5. The vector below represents the resultant of two forces acting on a particle P.

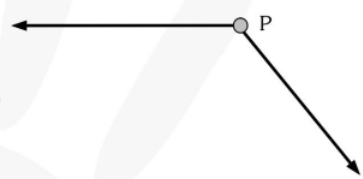


Which of the following pair of vectors best represents two forces that combine to produce this resultant force vector?

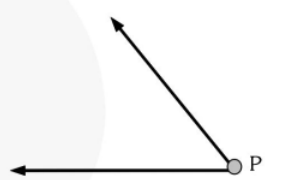
(A)



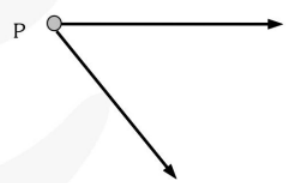
(B)



(C)



(D)



Q6. Two vectors $\vec{A}$ and $\vec{B}$ inclined at angle θ have a resultant $\vec{R}$ which makes an angle ϕ with $\vec{A}$.

If the directions of $\vec{A}$ and $\vec{B}$ are interchanged, the resultant will have the same :

(A) magnitude

(B) direction

(C) magnitude as well as direction

(D) neither magnitude nor direction

Q7. There are two force vectors, one of 5N and other of 12N at what angle the two vectors be added to get resultant vector of 17N, 7N and 13N respectively.

(A) 0°, 180° and 90° (B) 0°, 90° and 180°

(C) 0°, 90° and 90° (D) 180°, 0° and 90°

Q8. Following sets of three forces act on a body. Whose resultant cannot be zero ?

(A) 10, 10, 10 (B) 10, 10, 20

(C) 10, 20, 20 (D) 10, 20, 40

Unit Vector

Q9. $|\hat{i}|+|\hat{j}|-|\hat{k}| = ...$

(A) 3 (B) 2

(C) 1 (D) 0

Q10. If a unit vector is represented by $0.5\hat{i}-0.8\hat{j}+c\hat{k}$, then the value of 'c' is :-

(A) 1 (B) $\sqrt{0.11}$

(C) $\sqrt{0.01}$ (D) $\sqrt{0.39}$

Q11. If the angle between the unit vectors $\hat{a}$ and $\hat{b}$ is 60°, then $|\hat{a}-\hat{b}|$ is

(A) 0 (B) 1

(C) 2 (D) 4

Resolution of Vectors

Q12. A man rows a boat with a speed of 18 km/hr in northwest direction. The shoreline makes an angle of 15° south of west. Obtain the component of the velocity of the boat along the shoreline.

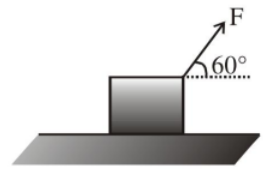
(A) 9 km/hr

(B) $18\frac{\sqrt{3}}{2}$ km/hr

(C) 18 cos (15°) km/hr

(D) 18 cos (75°) km/hr

Q13. A child pulls a box with a force of 200N at an angle of 60° above the horizontal. Then the horizontal and vertical components of the force are :



(A) 100N, 173.2 N

(B) 86.6N, 100 N

(C) 100N, 86.6 N

(D) 100N, 0 N

Expressing Vectors in Unit Vector Notation

Q14. The projection of a vector, $\vec{r} = 3\hat{i}+\hat{j}+2\hat{k}$, on the x–y plane has magnitude –

(A) 3 (B) 4

(C) $\sqrt{14}$ (D) $\sqrt{10}$

Q15. Given a vector $\vec{A} = 3\hat{i}+4\hat{j}-12\hat{k}$, then find out the direction cosines of this vector :-

(A) $\frac{-3}{7}, \frac{4}{7}, \frac{-12}{7}$ (B) $\frac{3}{16}, \frac{4}{16}, \frac{-12}{16}$

(C) $\frac{3}{5}, \frac{-4}{5}, \frac{-12}{5}$ (D) $\frac{3}{13}, \frac{4}{13}, \frac{-12}{13}$

Q16. A particle is situated at the origin of a coordinate system. The following four forces begin to act on the particle simultaneously.

$\vec{F}_1 = (2\hat{i}+3\hat{j}-\hat{k})$N, $\vec{F}_2 = (3\hat{i}-\hat{j}+\hat{k})$N;

$\vec{F}_3 = (2\hat{i}-5\hat{j}+3\hat{k})$N; $\vec{F}_4 = (5\hat{i}+3\hat{j}-3\hat{k})$N

The particle will move

(A) along x-axis (B) along y-axis

(C) in x-y plane (D) in y-z plane

Position and Displacement Vector

Q17. What is position vector of a particle situated at (50,-25,-10)?

(A) $50\hat{i}-25\hat{j}-9\hat{k}$ (B) $50\hat{i}+25\hat{j}-9\hat{k}$

(C) $50\hat{i}-25\hat{j}-10\hat{k}$ (D) $50\hat{i}+25\hat{j}-10\hat{k}$

Q18. A particle moves 14 m along the vector $6\hat{i}+2\hat{j}+3\hat{k}$, then 7 m along the vector $3\hat{i}-2\hat{j}+6\hat{k}$. Its total displacement (in meters) is -

(A) $15\hat{i}+2\hat{j}+12\hat{k}$

(B) $9\hat{i}+12\hat{k}$

(C) $9\hat{i}+6\hat{j}$

(D) $12\hat{i}+\hat{j}+12\hat{k}$

Scalar Product of Two Vectors

Q19. A force $\vec{F}=5\hat{i}+6\hat{j}+4\hat{k}$ acting on a body, produces a displacement $\vec{S}=6\hat{i}-5\hat{k}$. Work done by the force is :-

(A) 10 units (B) 18 units

(C) 11 units (D) 5 units

Q20. Which of the following is perpendicular to $\hat{i}-\hat{j}-\hat{k}$?

(A) $\hat{i}+\hat{j}+\hat{k}$

(B) $-\hat{i}+\hat{j}+\hat{k}$

(C) $\hat{i}+\hat{j}-\hat{k}$

(D) none of these

Q21. If $\vec{a}=2\hat{i}+3\hat{j}+4\hat{k}$ and $\vec{b}=4\hat{i}+3\hat{j}+2\hat{k}$ angle between $\vec{a}$ and $\vec{b}$ is :-

(A) $\cos^{-1}\left(\frac{25}{29}\right)$ (B) $\sin^{-1}\left(\frac{25}{29}\right)$

(C) $\tan^{-1}\left(\frac{25}{29}\right)$ (D) $\cot^{-1}\left(\frac{25}{29}\right)$

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Two vectors of the same physical quantity are unequal if

(A) They have different magnitudes and different directions

(B) They have different magnitudes and same directions

(C) They have same magnitudes and different directions

(D) They have same magnitudes and same directions

Q2. Which of the following statement(s) is/are true about a vector quantity :

(A) A physical quantity having magnitude & direction is a vector only if it obeys law of vector addition

(B) Two vector are equal if they have same direction and magnitude even if they represent two different physical quantities.

(C) A scalar and a vector quantity can be added through vector law of addition.

(D) Similar to scalar addition the order of the terms in vector addition does not affect the result.

Q3. If $\vec{a}$ is a vector and x is a non–zero scalar, then

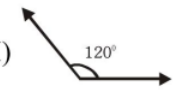
(A) $x\vec{a}$ is a vector in the direction of $\vec{a}$

(B) $x\vec{a}$ is a vector parallel to $\vec{a}$ if $x > 0$

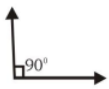
(C) $x\vec{a}$ and $\vec{a}$ can be antiparallel vectors

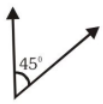
(D) $x\vec{a}$ and $\vec{a}$ can be equal vectors

Q4. Two displacement vectors ($\vec{P}$ and $\vec{Q}$) of same magnitude are arranged in the following manner

(I) 

(II) 

(III) 

(IV) 

Which of the following are correct regarding magnitude of resultant?

(A) I < II (B) II > III

(C) I > II > III (D) IV > I

Q5. The two vectors $\vec{A}$ & $\vec{B}$ are drawn from a common point and $\vec{C} = \vec{A} + \vec{B}$ then angle between $\vec{A}$ & $\vec{B}$ is -

(A) 90° if $C^2 = A^2 + B^2$

(B) greater than 90° if $C^2 < A^2 + B^2$

(C) greater than 90° if $C^2 > A^2 + B^2$

(D) less than 90° if $C^2 > A^2 + B^2$

Q6. Two forces having magnitudes 11 N and 15 N respectively are acting on a particle then the magnitude of the resultant of these two forces can be :-

(A) 5 N (B) 30 N

(C) 21.5 N (D) 25.5 N

Q7. If $\vec{A} + \vec{B}$ is a unit vector along x-axis and $\vec{A} = \hat{i} - \hat{j} + \hat{k}$, then what could be $\vec{B}$?

(A) $\hat{i} - \hat{k}$ (B) $\hat{j} - \hat{k}$

(C) $\hat{i} + \hat{j} + \hat{k}$ (D) $-2\hat{i} + \hat{j} - \hat{k}$

Q8. If $\vec{v}_1 + \vec{v}_2$ is $\perp$ to $\vec{v}_1 - \vec{v}_2$, then

(A) $\vec{v}_1$ is $\perp$ to $\vec{v}_2$

(B) $|\vec{v}_1| = |\vec{v}_2|$

(C) $\vec{v}_1$ is null vector

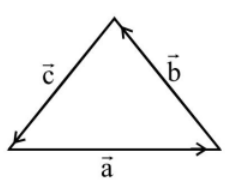
(D) None of these

MATCH THE COLUMN TYPE

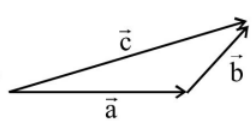
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9. Column-I show vector diagram relating three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$. Match the vector equation in column-II, with vector diagram in column-I :

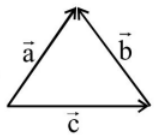
Column-I **Column-II**

(A) 

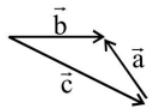
(P) $\vec{a} = \vec{b} + \vec{c}$

(B) 

(Q) $\vec{b} = \vec{c} + \vec{a}$

(C) 

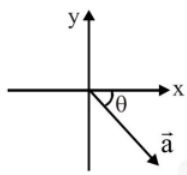
(R) $\vec{a}+\vec{b}+\vec{c}=0$

(D) 

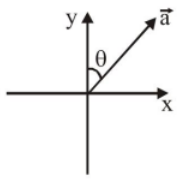
(S) $\vec{a}+\vec{b}=\vec{c}$

Q10. Match vector $\vec{a}$ shown in column-I with its unit vector representation in column-II.

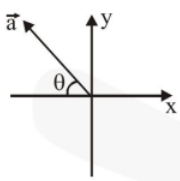
Column-I **Column-II**

(A) 

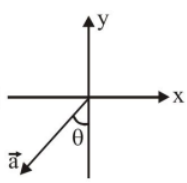
(P) $\vec{a}=a\sin\theta\,\hat{i}+a\cos\theta\,\hat{j}$

(B) 

(Q) $\vec{a}=-a\cos\theta\,\hat{i}+a\sin\theta\,\hat{j}$

(C) 

(R) $\vec{a}=-a\sin\theta\,\hat{i}-a\cos\theta\,\hat{j}$

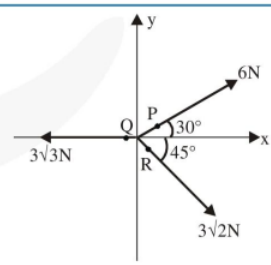
(D) 

(S) $\vec{a}=a\cos\theta\,\hat{i}-a\sin\theta\,\hat{j}$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q11. Three ants P, Q and R are pulling a grain with forces of magnitude 6N, $3\sqrt{3}$ N and $3\sqrt{2}$ N as shown in the figure. Find the magnitude of resultant force (in N) acting on the grain.



Q12. A bee is at position $\vec{r}_1=\left(2\hat{i}+4\hat{j}+3\hat{k}\right)$m at t = 0. It moves with a constant velocity $\vec{v}=\left(3\hat{i}+4\hat{j}\right)$m/s for two seconds. Final position of the bee is $\vec{r}_2=\left(a\hat{i}+b\hat{j}+c\hat{k}\right)$m at t = 2 s. Find the value of (b – a – c)

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Two vectors $\vec{a}$ and $\vec{b}$ add to give a resultant $\vec{c} = \vec{a} + \vec{b}$. In which of these cases angle between $\vec{a}$ and $\vec{b}$ is maximum: (a, b, c represent the magnitudes of respective vectors)

(A) $c = a + b$

(B) $c^2 = a^2 + b^2$

(C) $c = a - b$

(D) can not be determined

Q2. At what angle should the two forces 2P and $\sqrt{2}\,P$ act so that the resultant force is $P\sqrt{10}$:

(A) 45° (B) 60°

(C) 90° (D) 120°

Q3. The sum of magnitudes of two forces acting at a point is 16N. If the resultant force is 8N and its direction is perpendicular to smaller force, then the forces are :–

(A) 6N and 10N (B) 8N and 8N

(C) 4N and 12N (D) 2N and 14N

Q4. A bird moves from point (1 m, –2 m, 3 m) to (4 m, 2 m, 3 m). If the speed of the bird is 10 m/s, then the velocity vector of the bird in m/s is :

(A) $5\left(\hat{i} - 2\hat{j} + 3\hat{k}\right)$ (B) $5\left(4\hat{i} - 2\hat{j} + 3\hat{k}\right)$

(C) $0.6\hat{i} + 0.8\hat{j}$ (D) $6\hat{i} + 8\hat{j}$

Q5. A particle moves $4\sqrt{2}$ m in South–West direction, 12 m in upward direction and then 1m in East direction. Find the net displacement.

(A) 13 m (B) 17 m

(C) 14 m (D) 18 m

Q6. A vector of magnitude 10 m in the direction 37° south of west has its initial point at (5 m, 2 m). If positive x–axis represents the east and positive y–axis the north, the coordinates of its terminal point are

(A) (–3 m, –4 m) (B) (3 m, 4 m)

(C) (–4 m, 6 m) (D) (–4 m,–6 m)

Q7. Find the component of $\vec{r}$ in the direction of $\vec{a}$:–

(A) $\frac{(\vec{r}.\vec{a})\vec{a}}{a^2}$ (B) $\frac{(\vec{r}.\vec{a})\vec{a}}{a}$

(C) $\frac{(\vec{r}\cdot\vec{a})\hat{r}}{r}$ (D) $\frac{(\vec{r}\cdot\vec{a})\hat{r}}{r^2}$

Exercise - 2A — Brain Booster

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statements about the sum of the two vectors $\vec{A}$ and $\vec{B}$, is/are correct?

(A) $|\vec{A} + \vec{B}| \le A + B$

(B) $|\vec{A} + \vec{B}| \ge A + B$

(C) $|\vec{A} + \vec{B}| \ge |\vec{A} - \vec{B}|$

(D) $|\vec{A} + \vec{B}| \ge |A - B|$

Q2. The angle between $\vec{A}+\vec{B}$ and $\vec{A}-\vec{B}$ can be :-

(A) 60° (B) 90° (C) 120° (D) 150°

Q3. Ayushi says that the sum of two vectors by the parallelogram method is $\vec{R}=5\hat{i}$. Subhangi says it is $\vec{R}=\hat{i}+4\hat{j}$. Both used the parallelogram method, but one used the wrong diagonal. Which one of the vector pairs below contains the original two vectors?

(A) $\vec{A}=+3\hat{i}-2\hat{j}\,;\ \vec{B}=-2\hat{i}+2\hat{j}$

(B) $\vec{A}=-3\hat{i}-2\hat{j};\ \vec{B}=+2\hat{i}+2\hat{j}$

(C) $\vec{A}=+3\hat{i}+2\hat{j};\ \vec{B}=+2\hat{i}-2\hat{j}$

(D) $\vec{A}=+3\hat{i}+2\hat{j};\ \vec{B}=-2\hat{i}+2\hat{j}$

Q4. If $\vec{A}+\vec{B}+\vec{C}=\vec{0}$ and $|\vec{A}|=2\,;\ |\vec{B}|=4$, then which of the following may be **CORRECT**?

(A) $|\vec{C}|=2$ (B) $|\vec{C}|=6$

(C) $|\vec{C}|=8$ (D) $|\vec{C}|=4$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 5 to 7

A boy lost in a jungle finds a note. In the note was written the following things.

Displacements

1. 300 m 53° South of East.
2. 400 m 37° North of East
3. 500 m North
4. 500 √2 m North-West
5. 500 m South

He starts walking at constant speed 2 m/s following these displacements in the given order.

Q5. How far and in which direction is he from the starting point after 5 min. and 50 s?

(A) 500 m due East

(B) 500 m due West

(C) 700 m due South-West

(D) 700 m due North-East

Q6. How far and in which direction is he from the starting point after 10 minutes?

(A) 500 √2 m due North

(B) 1200 m due North-East

(C) 500√2 m due North-East

(D) 900 m due 37° North of East

Q7. How far and in which direction has he finally displaced after all the displacements in the note?

(A) 500√2 m due North-East

(B) 500 m due North

(C) 866 m due North-West

(D) 500√3 m due 60° North of West

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q8. $\vec{a}$ and $\vec{b}$ are unit vectors and angle between them is $\frac{\pi}{k}$. If $\vec{a}+2\vec{b}$ and $5\vec{a}-4\vec{b}$ are perpendicular to each other then find the integer value of k.

Answer Key

Ex-1											
1	2	3	4	5	6	7	8	9	10	11	12
D	A	D	A	C	A	A	D	C	B	B	A
13	14	15	16	17	18	19	20	21			
A	D	D	A	C	A	A	D	A			

Ex- 1A

1. A, B ,C **2. A, D** **3. B, C,D** **4. A, B, D** **5. A,B,D**

6. A,C,D **7. B,D** **8. B** **9. (A)-R, (B)-S, (C)-P, (D)-Q**

10. (A)-S, (B)-P, (C)-Q, (D)-R **11. 3** **12. 1**

Ex-2						
1	2	3	4	5	6	7
C	A	A	D	A	A	A

Ex- 2A

1. A,D **2. A,B,C,D** **3. C,D** **4. A,B,D** **5. A**

6. C **7. B** **8. 3**